# LEADZM Electronic Scale

## User Manual

Precision balance, 3000 g x 0.01 g

Please read this manual carefully before use and keep it for future reference.

The scale has been calibrated before leaving the factory. Do not start the calibration unless a reference weight of the correct value is available.

# 1  Overview

This precision scale weighs up to 3000 g in steps of 0.01 g, converts between eleven units, counts pieces and weighs in percent of a reference sample. Figure 1 shows the main parts; Figure 2 shows the control panel.

| Ref. | Part |
|---|---|
| A | Weighing pan |
| B | Pan surround |
| C | Control panel |
| D | Display |
| E | Power cord |

# 2 Control panel and keys

| No. | Key / part | Function |
|---|---|---|
| 1 | ON/OFF | Switches the scale on; press again to switch it off. |
| 2 | Display | Shows the weight, the active unit, the stability mark and the battery symbol. |
| 3 | TARE/ZERO | Sets the display to zero; subtracts the weight of a container. |
| 4 | UNIT | Steps through the weighing units. |
| 5 | PCS | Enters the piece-counting mode. |
| 6 | CAL | Starts the calibration with a reference weight. |

The display shows the weight with the active unit on its right-hand side (g, ct, oz, ozt, lb, TOL, dwt, mm, kg, pcs or %). A small circle in the upper left corner is the stability mark; the battery symbol in the lower right corner shows the battery state.

# 3 Specifications

| | |
|---|---|
| Maximum weighing | 3000 g |
| Resolution | 0.01 g |
| Tare range | 3000 g |
| Calibration weight | 2000 g |
| Weighing pan | Round, stainless steel, 130 mm |
| Units | g, ct, oz, ozt, lb, TOL, dwt, mm, kg, pcs, % |
| Power | AC adapter (DC 9 V) or batteries |
| Operating temperature | 5 to 35 C |
| Auto power-off | None: the scale stays on until switched off |

# 4 Preparation

- Place the scale on a stable, level surface away from draughts, vibration, direct sunlight and strong electromagnetic fields.
- Connect the power adapter, or insert batteries, and switch on. Let the scale warm up for a few minutes before precise measurements.
- Keep the pan clean and do not exceed the maximum capacity, container included.

# 5 Switching on and off

1. Press the ON/OFF key. The display shows all segments (8.8.8.8.8), then the maximum weighing value (3000), the software version (u-1908) and a row of dashes (-----) while the sensor settles.
2. When the display shows 0.00 g the scale is ready. The unit after switching on is always grams.
3. To switch off, press the ON/OFF key again. The scale does not switch off by itself.

# 6 Weighing

1. Make sure the display shows 0.00. If not, press the TARE/ZERO key with the pan empty.
2. Place the object on the centre of the pan and read the weight once the stability mark is shown.

## 7 Tare: weighing in a container

1. Place the empty container on the pan. Its weight is displayed.
2. Press the TARE/ZERO key. The display shows 0.00: the container weight is subtracted.
3. Put the material into the container. The display shows the net weight.
4. Pressing TARE/ZERO with the pan empty returns the display to 0.00.

## 8 Changing the unit

Press the UNIT key to step through the units. Each press moves one step along the sequence below and the unit indicator on the right of the display changes accordingly; after the last unit the scale returns to grams.

g → ct → oz → ozt → lb → TOL → dwt → mm → kg → % → g

- TOL is the tola, dwt the pennyweight, mm the momme.
- The % step is the percentage mode (section 10). Without a reference sample the display shows Err SP for a moment and then %.
- The unit returns to grams every time the scale is switched on.

## 9 Counting pieces (PCS)

1. Place the empty container, if one is used, on the pan and press TARE/ZERO so that the display shows 0.00.
2. Press the PCS key. The display shows -CoU- briefly, then a flashing 10, and the unit indicator changes to pcs.
3. Put 10 pieces of the parts to be counted on the pan and press the PCS key again. The display shows dashes while the average piece weight is determined, then 10.
4. Add the remaining pieces. The display shows the number of pieces.
5. To leave the counting mode press the PCS key once more. The display shows ----- and returns to grams. The UNIT key does not leave the counting mode.

If the PCS key is pressed to confirm while the pan is empty, the display shows Err-SP and the scale enters the counting mode without a reference; press PCS again to return to grams. The weight of one piece must be larger than the resolution of the scale, 0.01 g.

## 10 Percentage weighing

1. Place the reference sample on the pan and tare if a container is used.
2. Press the UNIT key until % is shown. The sample is taken as 100 %.
3. Replace the sample with the material to be compared; the display shows its weight in percent.

# 11 Calibration (CAL)

Calibrate only with a 2000 g reference weight, after a warm-up of 30 minutes.

1. With the pan empty, press and hold the CAL key for about 3 seconds until -CAL- is shown, then release.
2. The display flashes the value of the reference weight (2000). Place the 2000 g weight on the pan.
3. The display shows dashes, then the weight value. Remove the weight; the display shows dashes again, then 0.00. The calibration is complete.
4. If the readings are still inaccurate, repeat the procedure.

# 12 Error messages

| Display | Meaning | What to do |
|---|---|---|
| Upper horizontal line | The load on the pan is negative | Remove everything from the pan and press TARE/ZERO; recalibrate if it persists |
| Lower horizontal line or -------- | Overload | Remove the load; do not exceed 3000 g including the container |
| Err-1 | Switched off and on again too quickly | Wait 3 seconds before switching on |
| Err-2 | Weighing unstable | Wait a moment; avoid draughts and vibration |
| Err-SP | Counting or percentage mode entered without a reference sample | Counting: press PCS to return to grams. Percentage: place the reference sample first |
| Battery symbol flashing, then shut down | No power | Replace the batteries or check the adapter and cord |

# 13 Precautions

- Never exceed 3000 g on the pan, container included. The overload warning is a row of dashes.
- Press the keys gently and one at a time. Do not press on the pan with a hard object.
- To remove the round pan, turn it clockwise and lift. Do not pull it straight up.
- Keep the scale dry. Clean it with a soft, slightly damp cloth; do not use solvents.
- Wait at least 3 seconds between switching off and on again.
- Do not press the CAL key without a reference weight; the factory calibration would be lost.
- Store the scale in its original packaging, with the batteries removed, when it is not used for a long time.